Scientific Justification:

Maritime domain awareness — detecting dark vessels, AIS spoofing, and voyage gap-filling — requires the ability to re-identify individual ships across multiple satellite passes. We are developing a metric learning pipeline for SAR ship re-identification using MMSI identifiers from AIS as free ground truth, eliminating the need for manual labeling.

Using Sentinel-1 IW GRD (10m) over the Gulf of Mexico, we have assembled a dataset of 20,000+ ship chips across 4,100+ unique vessels (Aug–Dec 2024). Our baseline results show meaningful clustering for large vessels (≥150m): cargo ships achieve mean intra/inter ratio of 0.672, with individual vessels such as CAPE KENNEDY (212m) reaching 0.295. However, GRD resolution fundamentally limits discriminability — a 150m ship occupies only 15 pixels, insufficient to resolve deck structure, superstructure geometry, or individual scattering centers.

TerraSAR-X Spotlight mode (~1m) would provide 150 pixels for the same vessel, resolving hatches, cranes, deck equipment and hull geometry — features that are identity-discriminative. A direct comparison of GRD (10m) vs TerraSAR-X Spotlight (~1m) re-ID accuracy would quantify the information gain from resolution and establish the minimum resolution threshold for reliable open-water ship re-identification — a question with direct implications for the design of future maritime surveillance constellations.

Data Request:

- Mode: Spotlight (preferred) or StripMap
- Polarization: Dual-pol VV+VH
- Area: Gulf of Mexico open water, approx. lat 26–28.5°N, lon -96.5 to -88°W
- Scenes: 20–30 acquisitions, matched to dates with existing Sentinel-1 GRD coverage (Aug–Dec 2024)
- Format: SSC (Single-look Slant-range Complex)

Expected Outputs:

- Peer-reviewed publication comparing GRD vs SLC vs TerraSAR-X re-ID accuracy (IEEE GRSL or IEEE TGRS target)
- Public release of ship chip dataset and model weights on HuggingFace/Zenodo
- Submitted companion paper in IEEE GRSL: Few-Shot SAR ATR via Prototypical Networks (May 2026)

PI: Mousa Shaya, Independent Researcher — mshaya01.ml@gmail.com